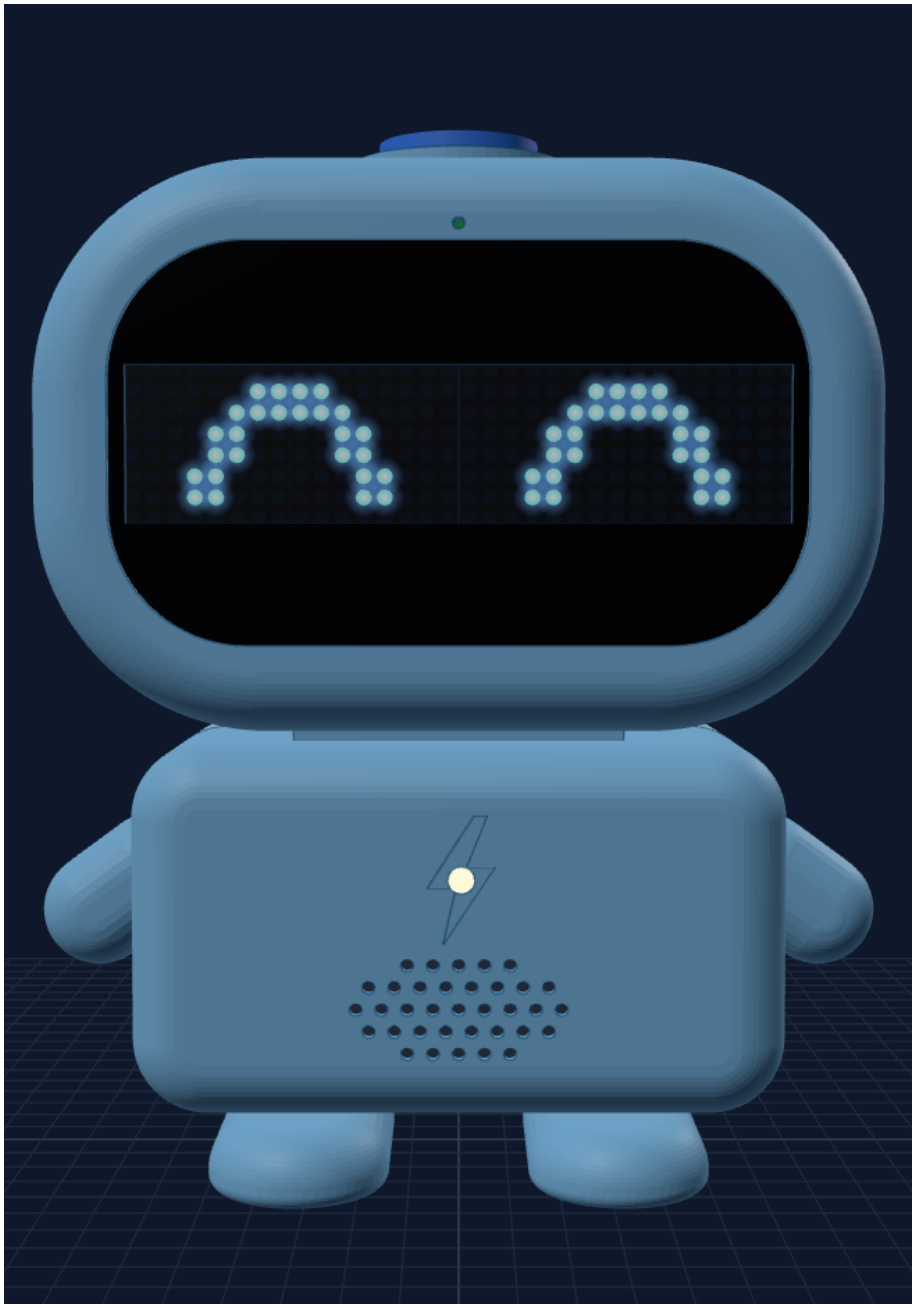
Sparky - build & print guide

Enclosure v6 ('little astronaut'), designed in SolidCraft. Written for a first-time 3D printer user. All sizes in millimetres.



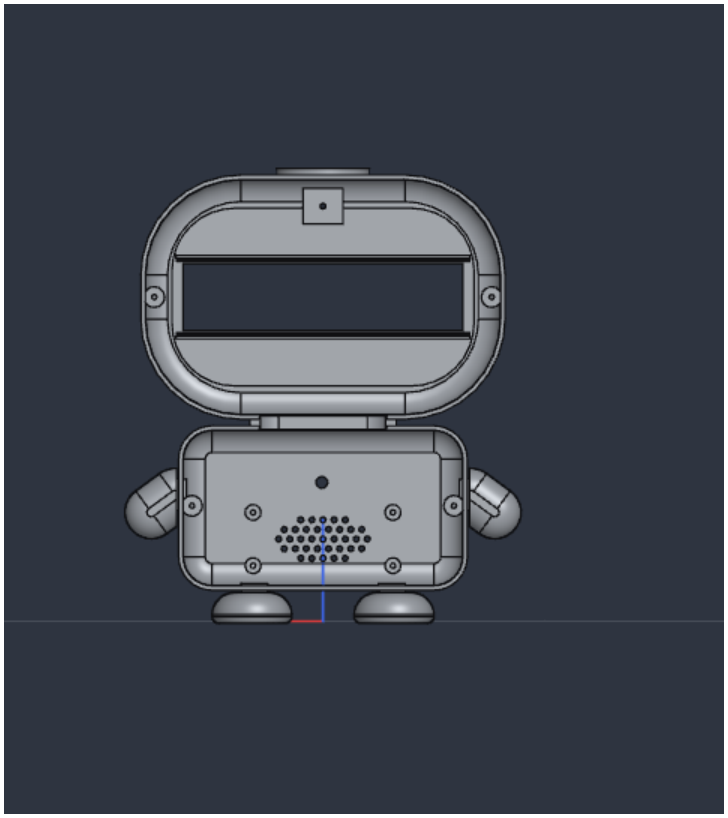
Sparky as designed, showing the 'happy ^ ^' face on his 32 x 8 blue LED matrix behind the black screen.

Short answer: yes, this design should work. Every component fits inside with the sizes from its datasheet, Sparky stands steadily on his feet, and he closes with four screws and no glue. It has not been printed yet, so the first print and a fit test with the real parts are still the real proof. The biggest risk is not the design but the print itself: the front piece is a large, tall part that needs supports.

1. What you are printing

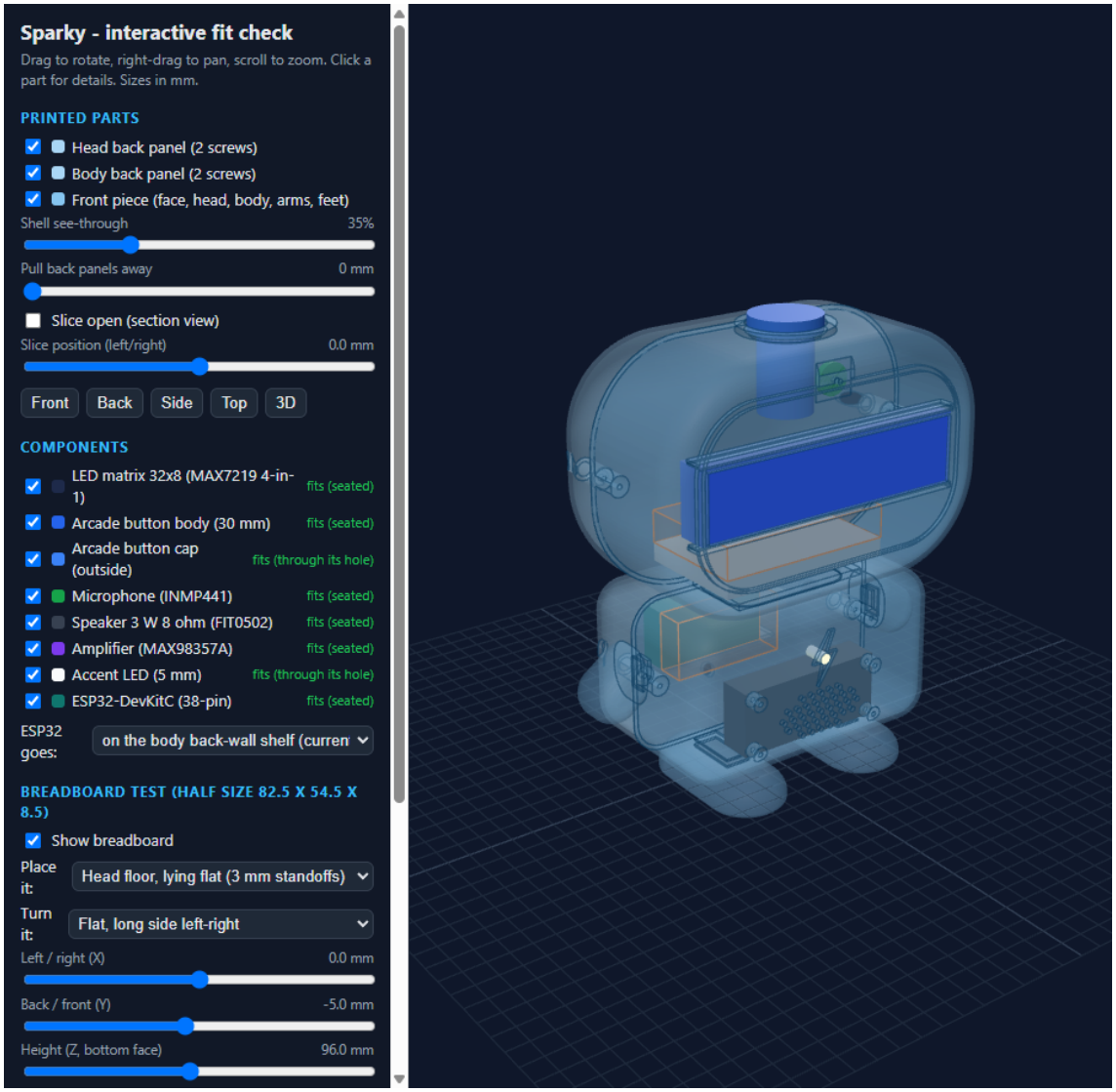
Sparky is three printed parts. The front piece is one solid print with the face, head, body, arms and feet already joined. The two back panels screw on so you can open him again for repairs.

Part	Size (approx.)	What it holds	How it attaches
Front piece	179 x 75 x 205	Matrix, button, mic, speaker, amp, LED - everything mounts here	-
Head back panel	164 x 48 x 110	Nothing (cover)	2 x M2.5 screws from the back
Body back panel	130 x 39 x 77	ESP32 shelf, USB cable exit	2 x M2.5 screws from the back



The front piece seen from the back (panels removed): matrix rails, mic pad, screw posts, speaker posts and the neck opening for wires.

2. Where every component goes - and does it fit?

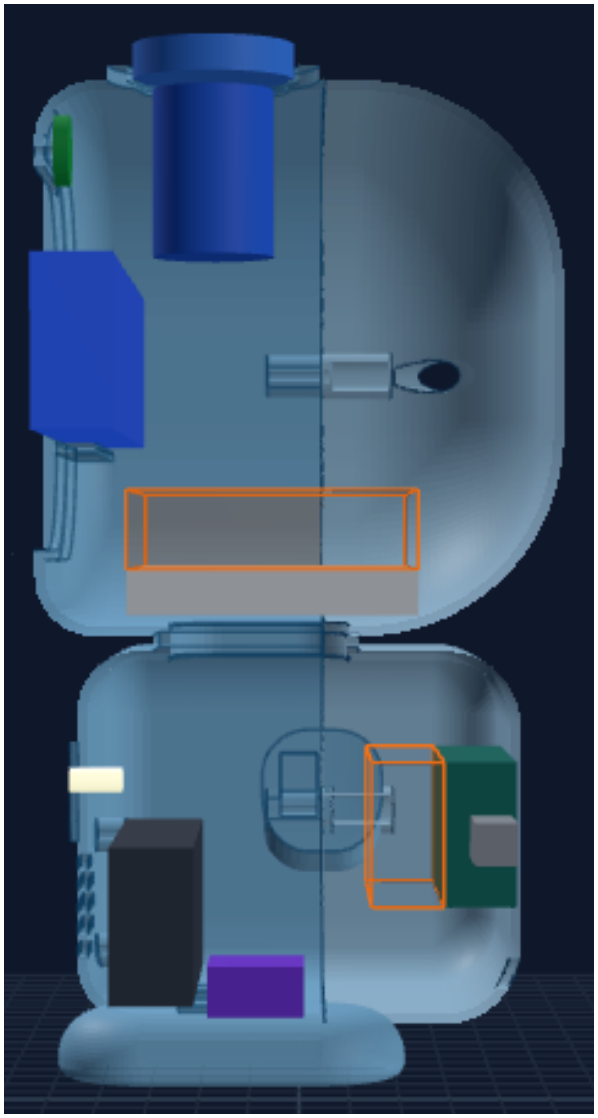


The interactive fit-check viewer (docs/sparky_3d_viewer.html): the shell is see-through, every component is in its planned place, and each one is checked against the walls.

Each component was modelled at its datasheet size and checked against the printed walls and against the other components. 'Fits' means nothing goes through a wall.

Component	Real size	Where it goes	Held by	Fit
LED matrix 32x8 (MAX7219 4-in-1)	128-129 x 32-33 x 12-15, ~55 g	Head, behind the face window	Two rails (33.6 mm slot)	Fits
Arcade button 30 mm	~33 deep under the panel	Top of the head, behind the matrix	Snaps into the 30 mm hole	Fits
Microphone INMP441	14 round	Behind the forehead hole	Flat pad - clip still to add	Fits
Speaker 3 W 8 ohm (FIT0502)	70 x 30 x 16, 24 g	Belly, behind the dot grille	4 screws on posts (63 x 24 pattern)	Fits
Amplifier MAX98357A	~19 x 18	Body floor, between the foot pegs	Clip/tape still to add	Fits
ESP32-DevKitC (38-pin)	54.4 x 27.9	Shelf on the body back wall	Shelf - clip still to add	Fits
Accent LED 5 mm	5 round	Through the chest lightning bolt	Push-fit	Fits
Half breadboard (optional)	82.5 x 54.5 x 8.5	Head floor, lying flat on 3 mm standoffs	Standoffs still to add	Fits (1-2 mm gap)

All components were checked with the rest of Sparky assembled. Positions are the planned ones; after your parts arrive, measure them and the clips can be added.

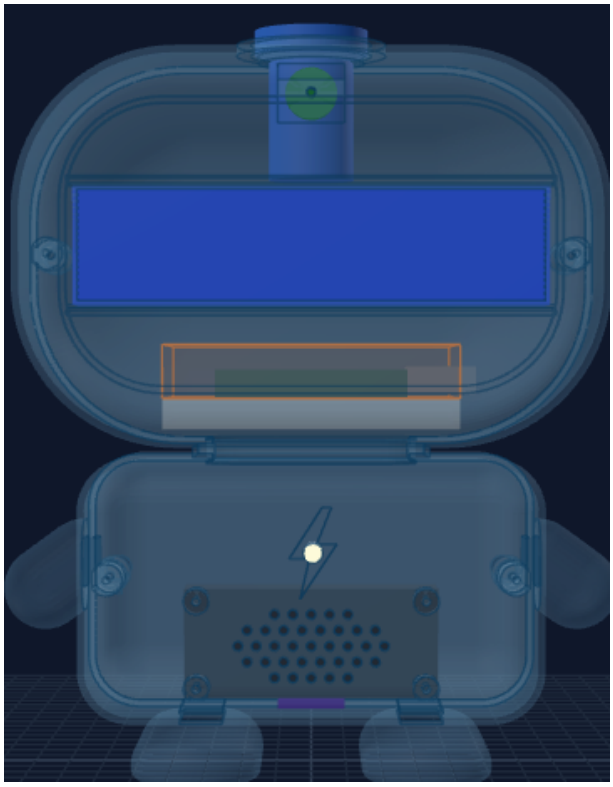


Cut through the middle (front on the left): matrix behind the face, button above it, breadboard on the head floor, speaker and amp in the belly, ESP32 on its back shelf.

3. The breadboard question

You asked whether the half breadboard could go inside. The fit check tried it in five places:

Where	Result
Head floor, lying flat (3 mm standoffs)	**Fits** - with a 1-2 mm gap to the walls, even with the ESP32 plugged into it, its USB plug, and 15 mm of room for jumper wires
Body floor, lying flat	Does not fit - hits the amplifier
Body back wall, standing up	Does not fit - goes through the rounded back wall
Head back, standing up	Does not fit - goes through the dome
Anywhere outside the head floor	Use the sliders in the viewer to try your own ideas



The breadboard (grey) with the ESP32 (green) on the head floor, under the LED matrix (blue). The orange box is room for jumper wires.

So it *can* live in the head. That is handy: the matrix, button and mic are all in the head, so their wires become very short, and only the speaker, amp, LED and USB wires go down through the neck. The USB cable then runs from the back of the body, up through the neck, to the ESP32. Two things to know: it is a tight fit (1-2 mm), and the breadboard needs two small standoffs on the head floor, which are not in the CAD yet. For a later, tidier build, a small soldered perfboard does the same job and is easier to fix in place.

4. Will he stand up?

	Value
Total weight with all parts	about 500 g
Centre of mass	centred left-right, 8 mm in front of the middle, 90 mm high
Tilt needed to fall backwards	about 19 degrees
Tilt needed to fall forwards	about 25 degrees
Tilt needed to fall sideways	about 29 degrees

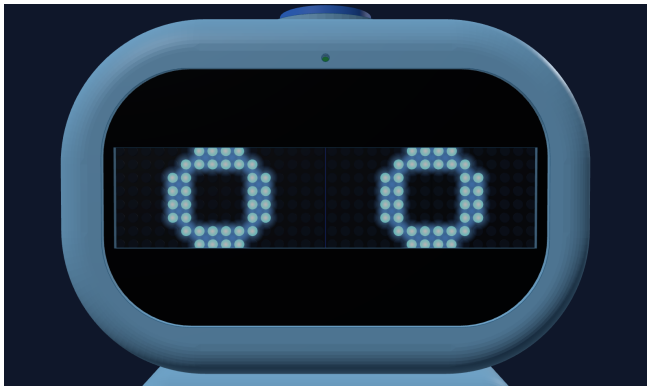
That is steady for a desk toy. The long feet (72 mm, reaching under the whole body) are what make this work - an earlier version with short feet would have tipped over backwards at about 6 degrees, for example from a tug on the USB cable.

5. Sparky's face

The face is the 32 x 8 blue LED matrix: four 8 x 8 blocks of 3 mm LEDs on a 4 mm pitch, seen through a 126 x 30 mm window in a black screen (paint the recess black, or use black film). A piece of smoked film over the window hides the unlit LEDs and makes the lit ones pop.



Happy ^ ^ (the default)



Surprised O O

The viewer has seven ready-made faces - happy, surprised, neutral (listening), wink, sleepy, thinking and error - and a grid where you can click LEDs on and off to design your own. It prints each face as 8 lines of 32 characters ('#' = on), ready to copy into the firmware later.

One thing the render shows: the window frame covers about 1 mm around the edge of the matrix so the board cannot fall out forwards. The very top and bottom rows of LEDs are therefore about two-thirds visible (you can see it on the round 'O' eyes). Faces that keep the outer rows empty, like the happy face, look perfect. If you want every row fully visible, the window can be opened up to the full 128 x 32 mm and the matrix held only by its rails and clips.

6. 3D printing it - a guide for your first print

Big idea: do not make Sparky your very first print. Print small test pieces first, check your parts fit, then print the big piece.

Step by step

1. **Find a printer and a helper.** Ask whether the university has a makerspace or lab technician. Someone who knows the printer is worth more than any setting in this guide.
2. **Install the slicer** for that printer (Bambu Studio, PrusaSlicer or OrcaSlicer). The slicer turns the 3D file into printer instructions and tells you the real print time and material.
3. **Print the test pieces first** (about 20 minutes each): a 30 mm button hole, a short piece of the matrix slot, one screw post and the LED hole. Check the button snaps in, the matrix slides in, an M2.5 screw grips, and the LED pushes through. If something is too tight, the CAD can be adjusted before the long print.
4. **Print the body back panel** (smallest, quickest) and check it against the real ESP32.
5. **Print the head back panel.**
6. **Print the front piece last**, standing upright on its feet.

Settings to start with

Setting	Use	Why
Material	PLA	Easiest to print, cheap, comes in light blue
Nozzle	0.4 mm (0.6 mm if available)	0.6 mm is roughly 30-40% faster and suits the 2 mm walls
Layer height	0.2 mm (0.28 mm for speed)	0.28 is faster but shows more lines on the round helmet
Walls / perimeters	5 walls with a 0.4 nozzle (3 with 0.6)	Makes the 2 mm walls fully solid and the screw posts strong
Infill	15%	Only the feet and arms are thick enough to need it
Supports	Tree supports, on build plate and on the model	Needed under the head, under the arms, inside the head ceiling and across the top of the face window
Brim	Yes, 5-8 mm	The front piece is tall and only touches the bed with its feet - a brim stops it from tipping over mid-print
Orientation	Front piece upright on its feet; panels lying on their open side	The open back lets you reach in and remove the supports

Time and material

- Plastic: about 390 g for the three parts plus 50-100 g of supports. One 1 kg spool is enough, with room for a reprint.
- Time: very roughly 12-16 hours on a fast modern printer, or 30+ hours on an older one. Your slicer gives the real number when you load the file.
- Faster printing: use a 0.6 mm nozzle, 0.28 mm layers, and print the two back panels on another printer at the same time. Keep the 2 mm walls - thinner walls save little time and weaken the screw posts.

Files: hardware/cad/sparky_v6_mascot/sparky_mascot.3mf (all three parts, for the slicer) and sparky_mascot.step (for CAD).

7. Wiring and power tips

- **Get it working on the breadboard on your desk first.** The demo is about Sparky working, not about the shell.
- **Inside Sparky** you have two options: (1) the breadboard on the head floor (it fits - see section 3), or (2) jumper wires straight from the ESP32 pins to each module, with Wago 221 lever connectors (no soldering) to share 5 V and GND.
- **LED matrix pins:** if its 5-pin plug points out sideways from the end of the board, bend the pins backwards or solder the wires on - a sideways plug is too long to fit across the head.

- **Power is the most common ESP32 problem.** The matrix at full brightness plus the speaker at full volume can draw more than a laptop USB port gives. The ESP32 then keeps restarting with a 'brownout detector triggered' message. Keep matrix brightness low (about 2-4 out of 15) and use a 5 V / 2 A phone charger instead of a laptop port.
- **USB cable:** put a cable tie around the cable just inside the hole in the back. If someone pulls the cable, the tie takes the force instead of the ESP32's USB socket.

8. Putting him together (no glue)

1. Clean the supports out of all three parts.
2. Push the arcade button into the hole on top of the head until it clicks.
3. Slide the LED matrix into its rails from the back, LEDs facing the window.
4. Place the mic board on its pad behind the forehead hole, sound hole facing out.
5. Screw the speaker onto its four posts behind the grille (M2.5 or M3 self-tapping screws).
6. Push the 5 mm LED into the hole in the lightning bolt.
7. Place the amp on the body floor and the ESP32 on its shelf in the body back panel (or the breadboard on the head floor).
8. Feed the USB cable in through the hole at the back of the body and plug it into the ESP32.
9. Connect the wires; run the speaker/amp/LED wires up through the neck if the ESP32 is in the head.
10. Screw on the body back panel (2 screws), then the head back panel (2 screws).

Still to design once your parts arrive: small clips for the matrix (to stop it sliding backwards), the mic, the amp and the ESP32, plus breadboard standoffs if you choose the head position. They are left out until the real boards can be measured.

9. Plan up to the demo (28 October)

When	What
This week	Wire everything on the breadboard and test it. Print the test pieces.
When the parts arrive	Measure the matrix, ESP32 and mic boards so the clips can be added. Check which way the matrix pins point.
After the test pieces fit	Print the back panels, then the front piece.
Backup for the demo	If the print is not ready, demo with the open breadboard or the cereal-box mock-up - the electronics and software are what is being demonstrated.

10. Using the interactive 3D viewer

Open **docs/sparky_3d_viewer.html** in Chrome, Edge or Firefox. It works offline.

- Drag to rotate, right-drag to move, scroll to zoom. Click any part to see its name and size.
- 'Shell see-through' makes the walls transparent; 'Slice open' cuts Sparky in half so you can look inside.
- Untick the back panels, or slide 'Pull back panels away', to see how he opens.
- Breadboard test: choose a place and orientation, or move it with the sliders. The box underneath tells you straight away whether it fits, is tight, or hits a wall or another part.
- 'ESP32 goes' lets you put the ESP32 on its shelf or on the breadboard.
- The check uses each part's outer box, so near curved walls it errs on the safe side.

Sources for component sizes

Speaker FIT0502: robotshop.com (DFRobot Stereo Enclosed Speaker 3W 8 ohm), digikey.in (FIT0502), element14 (FIT0502), bastelgarage.ch (3W 8 ohm speaker in housing). LED matrix MAX7219 4-in-1: handsontec.com datasheet, beemong.com, diyables.io (FC16), makerguides.com, opencircuit.shop. ESP32-DevKitC and half breadboard: standard published board sizes.